

Lessons

1 Handling Missing Values

Drop missing values, or fill them in with an automated workflow.

2 Scaling and Normalization

Transform numeric variables to have helpful properties.

3 Parsing Dates

Help Python recognize dates as composed of day, month, and year.

4 Character Encodings

Avoid UnicodeDecodeErrors when loading CSV files.

5 Inconsistent Data Entry

Efficiently fix typos in your data.

FileEditSelectionViewGo

Query_processing

unit1_8.pyunit1_7.pyunit1_6.pyunit1_4.pyunit1_3.pysqlldbData_Cleaning.ipynbdata_cleaning.csvData_Cleaning.ipynbdata_cleaning.csvPython 3.11.9

EXPLORER

QUERY_PROCESSING

app.pyconnecting_sql_in_py.ipynbdata_cleaning.csvData_Cleaning.ipynbexample.csvmarket_data.csvmarket_data.xlsxnull_dataset.pysqlldbunit1_1.pyunit1_2.pyunit1_3.pyunit1_4.pyunit1_5.pyunit1_6.py

PROBLEMS

Filter (e.g. text, **/*.ts, !**/node_modules/...)

No problems have been detected in the workspace.

Data_Cleaning.ipynb

df = pd.read_csv("C:/Documents/Query_processing/data_cleaning.csv", encoding="utf-8")

+ Code+ MarkdownRun AllRestartClear All OutputsJupyter VariablesOutline

modules we'll useimport pandas as pdimport numpy as np

read in all our data

df = pd.read_csv("data_cleaning.csv")

print(df.head(10))

set seed for reproducibilitynp.random.seed(0)

[?]0.0s

Employee_IDNameAgeSalaryJoining_DateDepartmentCity

0101John25.035000.02022-01-12HRChennai

1102Anita28.042000.02022-02-15ITBangalore

2103Rahul31.052000.02022-03-20FinanceMumbai

3104Sneha29.048000.02022-04-25SalesDelhi

4105Karan35.061000.02022-05-18MarketingHyderabad

5106Priya27.043000.02022-06-10HRPune

6107Arun30.055000.02022-07-15ITKolkata

7108Meena32.060000.02022-08-08FinanceChennai

8109Ravi26.039000.02022-09-11SalesMadurai

9110Asha24.036000.02022-10-12MarketingCoimbatore

Cleaning Techniques

1. Missing Values

markdown

Fill numeric columns

The screenshot displays a Jupyter Notebook environment. The left sidebar shows the Explorer pane with a file tree containing various Python files and notebooks. The main workspace shows a code cell titled 'Cleaning Techniques' with the following code:

```

# Fill numeric columns
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Salary"] = df["Salary"].fillna(df["Salary"].median())

# Fill categorical columns
df["Department"] = df["Department"].fillna(df["Department"].mode()[0])
df["Name"] = df["Name"].fillna("Unknown")

# Fill dates
df["Joining_Date"] = df["Joining_Date"].fillna(method="ffill")
df.head(10)

```

Below the code, a warning message is shown: `FutureWarning: Series.fillna with 'method' is deprecated and will raise in a future version. Use df['Joining_Date'].fillna(method='ffill')`. The output of the code is a preview of the DataFrame, showing the first 10 rows of data.

	Employee_ID	Name	Age	Salary	Joining_Date	Department	City
0	101	John	25.0	35000.0	2022-01-12	HR	Chennai
1	102	Anita	28.0	42000.0	2022-02-15	IT	Bangalore
2	103	Rahul	31.0	52000.0	2022-03-20	Finance	Mumbai
3	104	Sneha	29.0	48000.0	2022-04-25	Sales	Delhi
4	105	Karan	35.0	61000.0	2022-05-18	Marketing	Hyderabad
5	106	Priya	27.0	43000.0	2022-06-10	HR	Pune
6	107	Arun	30.0	55000.0	2022-07-15	IT	Kolkata

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Query_processing

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EXPLORER

QUERY_PROCESSING

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PROBLEMS

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No problems have been detected in the workspace.

2. Scaling & Normalization

from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

df["Salary"] = scaler.fit_transform(df[["Salary"]])

df.head(10)

[10] ✓ 0.0s

	Employee_ID	Name	Age	Salary	Joining_Date	Department	City
0	101	John	25.0	0.070175	2022-01-12	HR	Chennai
1	102	Anita	28.0	0.094737	2022-02-15	IT	Bangalore
2	103	Rahul	31.0	0.129825	2022-03-20	Finance	Mumbai
3	104	Sneha	29.0	0.115789	2022-04-25	Sales	Delhi
4	105	Karan	35.0	0.161404	2022-05-18	Marketing	Hyderabad
5	106	Priya	27.0	0.098246	2022-06-10	HR	Pune
6	107	Arun	30.0	0.140351	2022-07-15	IT	Kolkata
7	108	Meena	32.0	0.157895	2022-08-08	Finance	Chennai
8	109	Ravi	26.0	0.084211	2022-09-11	Sales	Madurai
9	110	Asha	24.0	0.073684	2022-10-12	Marketing	Coimbatore

3. Date Cleaning

+ Code+ Markdown

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File Edit Selection View Go

Query_processing

unit1_8.py unit1_7.py unit1_6.py unit1_4.py unit1_3.py sql.db Data_Cleaning.ipynb data_cleaning.csv

Data_Cleaning.ipynb > df = pd.read_csv("C:\Documents\Query_processing\data_cleaning.csv", encoding="utf-8")

+ Code + Markdown | Run All Restart Clear All Outputs Jupyter Variables Outline Python 3.11.9

EXPLORER

QUERY_PROCESSING

app.py connecting_sql_in.py.ipynb data_cleaning.csv Data_Cleaning.ipynb example.csv market_data.csv market_data.xlsx null_dataset.py sql.db unit1_1.py unit1_2.py unit1_3.py unit1_4.py unit1_5.py unit1_6.py

PROBLEMS

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No problems have been detected in the workspace.

3. Date Cleaning

```
df["Joining_Date"] = pd.to_datetime(
    df["Joining_Date"],
    format="mixed",
    errors="coerce"
)

#Replace invalid dates

df["Joining_Date"] = df["Joining_Date"].fillna(df["Joining_Date"].mode()[0])
df.head(10)
```

[18] ✓ 0.0s Python

	Employee_ID	Name	Age	Salary	Joining_Date	Department	City
0	101	John	25.0	0.070175	2022-01-12	HR	Chennai
1	102	Anita	28.0	0.094737	2022-02-15	IT	Bangalore
2	103	Rahul	31.0	0.129825	2022-03-20	Finance	Mumbai
3	104	Sneha	29.0	0.115789	2022-04-25	Sales	Delhi
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Ln 11, Col 12 Spaces: 2 Spaces: 4 CRLF Cell 9 of 11 Go Live

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Query_processing

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Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline Python 3.11.9

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PROBLEMS

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4. Character Encoding

```
df = pd.read_csv("C:\Documents\Query_processing\data_cleaning.csv", encoding="utf-8")

df = pd.read_csv("C:\Documents\Query_processing\data_cleaning.csv", encoding="latin1")

#Remove special characters

df["Name"] = (
    df["Name"]
    .str.encode("ascii", errors="ignore")
    .str.decode("ascii")
)

df.head(10)
```

[19] ✓ 0.0s Python

	Employee_ID	Name	Age	Salary	Joining_Date	Department	City
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3	104	Sneha	29.0	48000.0	2022-04-25	Sales	Delhi
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5	106	Priya	27.0	43000.0	2022-06-10	HR	Pune
6	107	Arun	30.0	55000.0	2022-07-15	IT	Kolkata
7	108	Meena	32.0	60000.0	2022-08-08	Finance	Chennai
8	109	Ravi	26.0	39000.0	2022-09-11	Sales	Madurai
9	110	Asha	24.0	36000.0	2022-10-12	Marketing	Coimbatore

Sakthi postgres 0 files and 1 cell to analyze

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Windows Taskbar

02:18 PM 20-07-2026

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Query_processing

unit1_8.pyunit1_7.pyunit1_6.pyunit1_4.pyunit1_3.pysql.dbData_Cleaning.ipynbdata_cleaning.csv

unit1_8.pyunit1_7.pyunit1_6.pyunit1_4.pyunit1_3.pyunit1_2.pyunit1_1.pyunit1_5.pyunit1_6.py

EXPLORER

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Data_Cleaning.ipynb > df = pd.read_csv("C:/Documents/Query_processing/data_cleaning.csv", encoding="utf-8")

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Python 3.11.9

5. Inconsistent Data Entry

markdown

```
df["Department"] = df["Department"].str.strip()

#Convert to proper case
df["Department"] = df["Department"].str.title()

#Replace abbreviations
df["Department"] = df["Department"].replace({
    "Fin.": "Finance",
    "Hr": "HR",
    "Hr": "HR",
    "hr": "HR",
    "hR": "HR",
    "It": "IT",
    "it": "IT"
})

#Remove duplicate rows
df = df.drop_duplicates()

#Remove duplicate employee IDs
df = df.drop_duplicates(subset="Employee_ID")

df
```

[14] ✓ 0.0s

Sakthi postgres

0 files and 1 cell to analyze

Spaces: 2 CRLF

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Query_processing

unit1_8.pyunit1_7.pyunit1_6.pyunit1_4.pyunit1_3.py

sqlldb

Data_Cleaning.ipynbXdata_cleaning.csv

EXPLORER

QUERY_PROCESSING

app.pyconnecting_sql_Lin_py.ipynbdata_cleaning.csvData_Cleaning.ipynbexample.csvmarket_data.csvmarket_data.xlsxnull_dataset.pysqlldbunit1_1.pyunit1_2.pyunit1_3.pyunit1_4.pyunit1_5.pyunit1_6.py

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+ Code + Markdown | ▶ Run All ⌂ Restart 🗑 Clear All Outputs | 📄 Jupyter Variables 📄 Outline ...

Python 3.11.9

df = df.drop_duplicates(subset="Employee_ID")

df

[14] ✓ 0.0s

Python

	Employee_ID	Name	Age	Salary	Joining_Date	Department	City
0	101	John	25.0	35000.0	2022-01-12	HR	Chennai
1	102	Anita	28.0	42000.0	2022-02-15	IT	Bangalore
2	103	Rahul	31.0	52000.0	2022-03-20	Finance	Mumbai
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5	106	Priya	27.0	43000.0	2022-06-10	HR	Pune
6	107	Arun	30.0	55000.0	2022-07-15	IT	Kolkata
7	108	Meena	32.0	60000.0	2022-08-08	Finance	Chennai
8	109	Ravi	26.0	39000.0	2022-09-11	Sales	Madurai
9	110	Asha	24.0	36000.0	2022-10-12	Marketing	Coimbatore
10	111	NaN	29.0	47000.0	2022-11-15	HR	Salem
11	112	Vikram	NaN	51000.0	2022-12-18	IT	Erode
12	113	Latha	34.0	NaN	2023-01-10	Finance	Trichy
13	114	NaN	31.0	58000.0	2023-02-12	Sales	Vellore
14	115	Deepak	27.0	NaN	2023-03-14	Marketing	Tirunelveli
15	116	Divya	NaN	49000.0	2023-04-15	HR	Chennai
16	117	NaN	33.0	62000.0	2023-05-16	IT	Bangalore
17	118	Naveen	28.0	NaN	2023-06-17	Finance	Mumbai
18	119	Krishna	NaN	53000.0	2023-07-18	Sales	Delhi
19	120	NaN	26.0	45000.0	2023-08-19	Marketing	Pune

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Sakthi postgres 0 files and 1 cell to analyze

Spaces: 2 CRLF Cell 9 of 11 Go Live